



PT 335-A Plus Multi OPERATION MANUAL

755 Griffith Court, Burlington, Ontario, Canada L7L 5R9
Tel: +(905) 633-7663 Fax: +(905) 637-4419 Toll Free: 1-800-265-3376
www.graphicwhizard.com

Table of Contents

Introduction	2
Important Information.....	2
Safety Precautions	3
Power Supply	3
Specifications.....	4
Unpacking and Installation	5
Set Up	5
Touch Screen Installation	6
Parts Assembly.....	8
Operation	11
Creasing/Perforating.....	11
Crease/Perf Depth.....	12
Outfeed Accessories.....	14
Air Vent Control	17
Pile Height Control.....	19
Sheet Separator Adjustment.....	20
Skew Adjustment.....	20
Programming.....	21
Job Set Up.....	25
Troubleshooting	30
Touch Screen Calibration	33
Error Codes	34
Maintenance	36

Introduction

The PT 335-A Plus Multi brings together all the same features and adaptability as the PT 335-B Multi and much more. Added strike perforation capabilities mean the PT 335-A Plus Multi can complete partial perforations in either direction in a single pass. It features a onboard 6" color touch screen with job creator software that provides quick and simple job changeover to maximize efficiency with less operator involvement. With speeds up to 8,040 sheets per hour, 64 job memory, and a proven 4" pile feeder, the PT 335-A Plus Multi can quickly execute a variety of jobs.

Important Information

This manual is specifically designed to help you operate the PT 335-A Plus Multi. Please read, understand, and keep this manual in a convenient place for easy access.

DO NOT operate the PT 335-A Plus Multi until you have read and understand the guidelines in this manual. If you have any questions, contact your local dealer or Graphic Whizard Inc. before operating this machine.

Graphic Whizard Inc. shall not be held liable for consequential damages or incidental consequential damages resulting from: improper or inadequate maintenance by the customer; unauthorized modifications or misuse; or operation outside of the environment specified for the machine.

Graphic Whizard Inc. is committed to continually improving the design and performance of the product line. Please note: the design and specifications of this machine are subject to change without notice and without legal obligation.

All rights are reserved. No part of this manual may be photocopied, reproduced, or translated to another language without the prior written consent of Graphic Whizard Inc.

Safety Precautions

- Please turn off and disconnect the power to the machine prior to performing maintenance. Make note of connections before disconnecting. Ensure all cables are properly secured to the board.
- Read this entire manual before you run the machine.
- Ensure all accessories are installed properly before running the machine.
- Ensure the electrical input and ground are accurate before powering on the PT 335-A Plus Multi. This machine operates on 110 volts
- Ensure all safety covers are in place. **DO NOT** run the PT 335-A Plus Multi without the covers installed. The top lid has an interlock switch that will stop the machine if open or removed.
- For your personal safety, disconnect power before cleaning the interior of the machine.
- Disconnect the power cord if the machine is not going to be used for an extended period.
- Install the PT 335-A Plus Multi on a level surface.
- **NEVER** operate the unit with wet hands or connect or disconnect the power cord when wet or when standing in water.
- Keep hair and clothing away from the unit.
- Keep hands clear of the unit when operating. Keep liquids away from the PT 335-A Plus Multi.
- **DO NOT** place any heavy material on the machine.
- To prevent the risk of shock or fire, keep metal and flammable materials out of the machine. If the unit is exposed to shock or fire and if it is safe to do so, power off and unplug the unit then call a service technician. Do not power on the unit without consulting a service technician.
- If the machine overheats or if you detect an odor or see smoke, immediately shut down the unit and contact your local service technician.

Power Supply

- The AC input voltage frequency is AC110/50-60Hz. Please use a power cord rated for 15A or higher.
- If the voltage or frequency is not within range, the machine may fail to function.
- Do not remove the power cord while machine is in use. Power down the PT 335-A Plus Multi before disconnecting the power cord.
- Ensure the unit is connected to a properly grounded outlet before powering on.

Specifications

Item	Specification
Electrical	110-130v, 50/60hz, 15a
Speed	8,040 sheets per hour (8.5" x 11", 1 crease)
Maximum Stock Size	13" x 35.4" (330.2 x 899.1 mm)
Minimum Stock Size	5.5" x 8.26" (139.7 x 209.8 mm)
Stock Weight	150- 400 GSM
Stock Type	Non-Coated, Coated, Laminated
Tabletop/Floor Model	Floor
Maximum # Crease Positions	Up to 32 creases/perfs per sheet
Program Memory	64 custom
Feeder	4" air fed vacuum pile feeder
Options	Linear multi tools Cross perforator Additional crease widths Extension table Narrow feeder (min- 1.77" x 5.31")
Linear Multi Tool Options	12, 24 or 48 TPI perf tools Strike perf option (up to 4 linear strike perfs) Linear score Slit Kiss cut
Crease/Perf Depth Control	Adjustable
Crease Die Options	0.6 mm 1 mm 1.2 mm 1.4 mm
Perf Die Options	12, 24 or 52 TPI

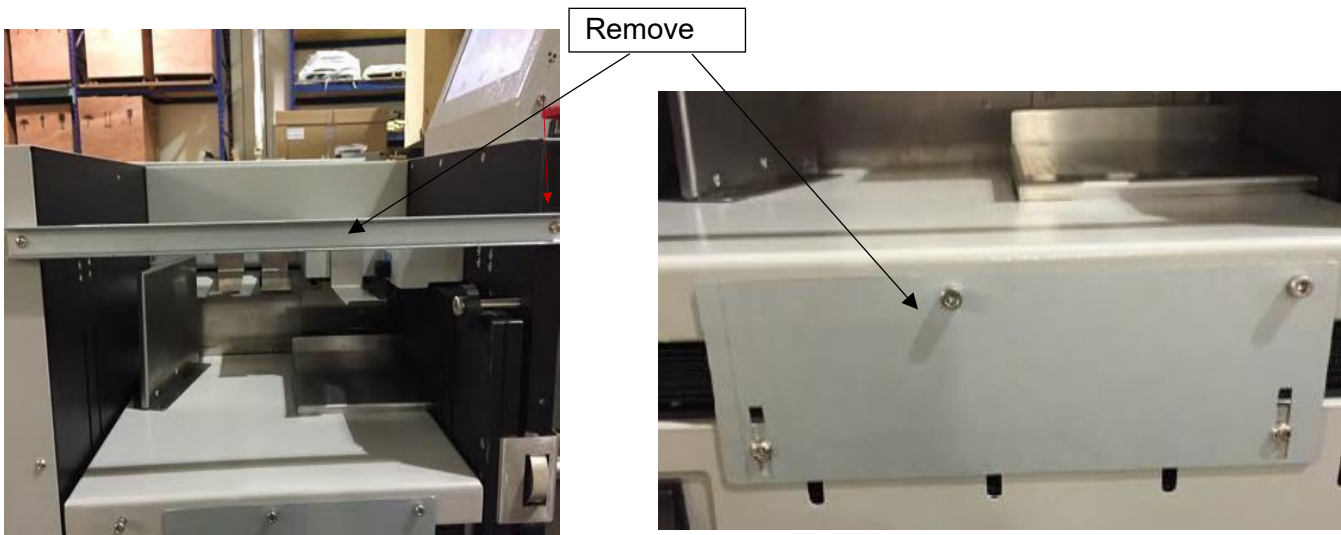
Unpacking and Installation

Set Up

Once the machine is unpacked, please ensure that it is positioned on a flat surface away from any potentially dangerous or otherwise harmful areas. The machine should sit on the stand in the position shown below.



Ensure all brackets are removed for shipping purposes (see images below). There should not be any brackets installed holding the feed table down once set up begins. During shipping there will be a support bar installed across the feeding table, as well as a small bracket holding the table down. Please remove both.



Touch Screen Installation

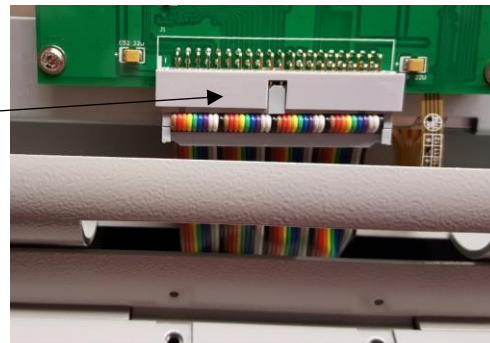
To prevent damage during shipping, the touch screen has been removed and placed in the bin below the machine along with other components. To install the touch screen on the PT 335-A Plus Multi, complete the following:

1. Remove the screws that hold the screen secure in the bracket.



2. Remove the four small screws securing the back cover and remove the back cover.
3. Remove the two screws in the base of the bracket. These screws are used to secure the touch screen assembly to the PT 335-A Plus Multi back cover.

4. Hold the touch screen over the PT 335-A Plus Multi cover and connect the ribbon cable by feeding it through the access hole. Ensure the cable is secure in the plug.





5. Secure the touch screen to the PT 335-A Plus Multi by slipping the foot of the bracket under the back cover then moving it forward until the screw holes are aligned with the holes in the cover. Install the screws. Replace the back cover.

6. Align the touch screen within the bracket for the optimum viewing angle and secure with the two screws. The screen is now installed.

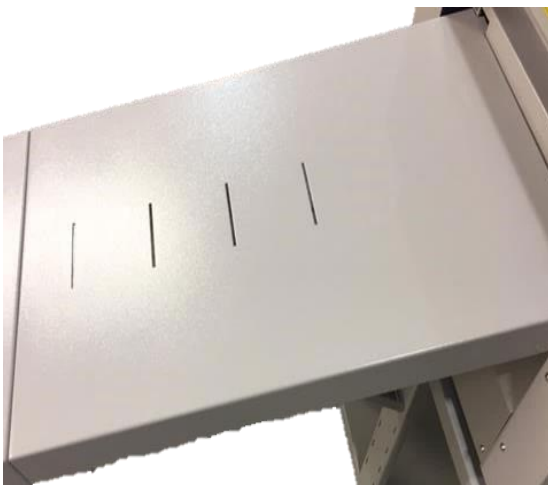
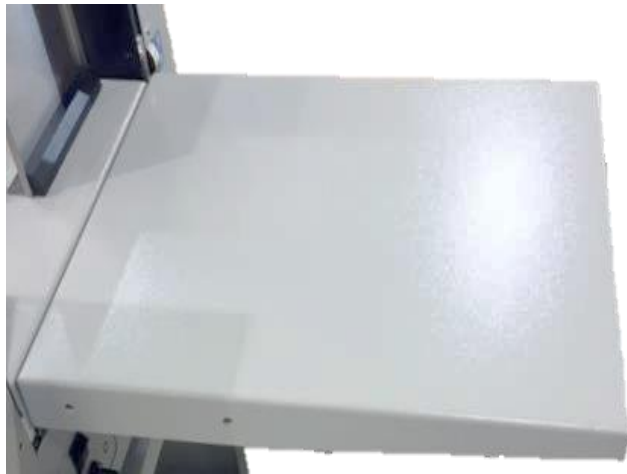


Parts Assembly

The PT 335-A Plus Multi arrives with the following parts:

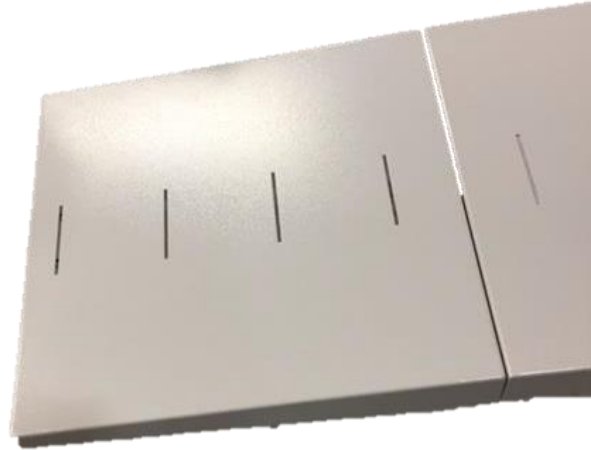
- Feed table extension
- Exit tray
- Exit tray extension
- Exit tray stops
- Side guides
- Back stops
- Paper feed guides (pillars)
- Paper deflectors

The feed table extension attaches to the three hex screws on the edge of the feed table. This is used for longer sheets.



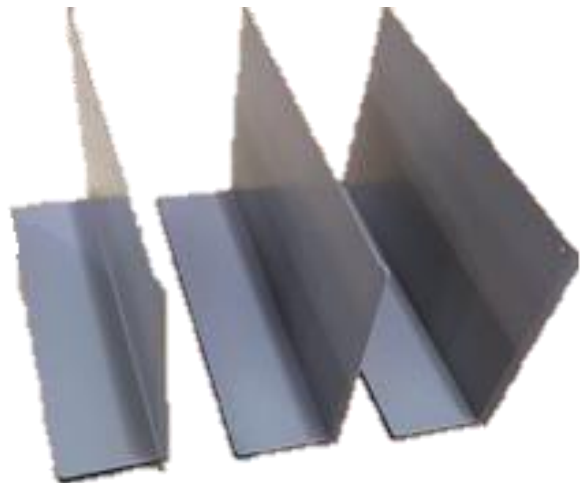
The exit tray hooks into the two slits at the back of the machine, while the support arm on the tray hooks into the desired slots on the stand.

The exit tray extension attaches to the three hex screws on the edge of the exit tray.



The exit tray back stops allow the user to stop pages from falling off the end of the exit tray.

The side guides help guide the finished product on the exit tray until it reaches the exit tray back stop.

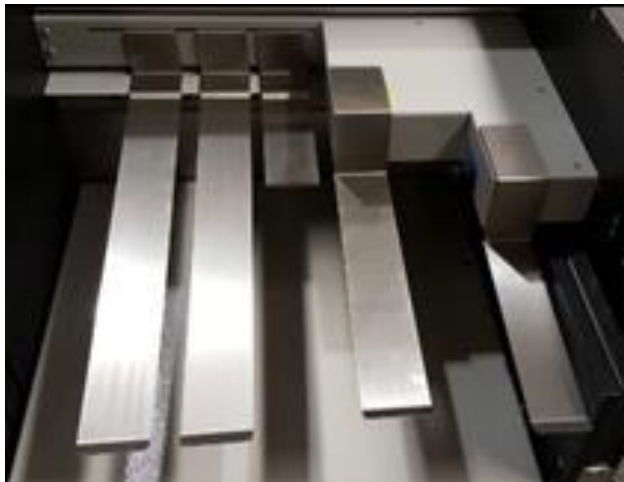




The paper guides (pillars) and back stop work together to keep the paper aligned and from slipping out of the feeder.

The paper deflectors help keep the paper in line with the feed entrance and are used when the stock is light or if the feeder is struggling to feed.

Note: it is not necessary to use all deflectors all the time. Add or remove deflectors to suit the job.



Operation

Creasing/Perforating

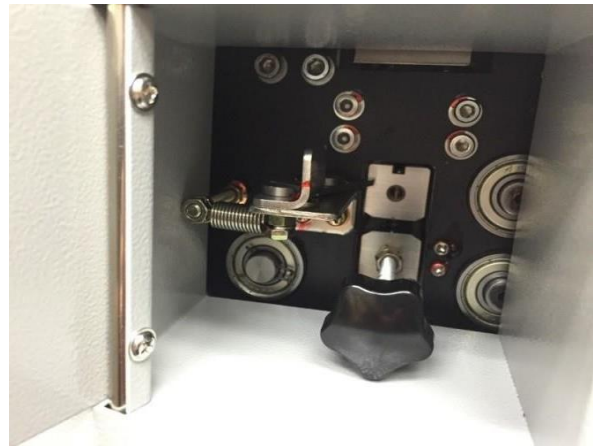
The PT 335-A Plus Multi is equipped with the PT exchangeable die system for quick exchange of creasing and perforating tools.

Crease tools: 1.0mm version is standard. Optional: 0.4mm, 0.6mm, 1.2mm, 1.4mm.

Perforate tools: 12 TPI (*Tooth-Per-Inch*), 24 TPI and 52 TPI versions all are optional.

To exchange the dies please follow the instructions below.

Shown here is a crease die in place with the safety latch engaged.



Utilizing one finger push the top of the safety latch to the side to disengage the lock (shown here). This will allow the operator to slide out the existing die and replace it.

While sliding in a new die hold the top and bottom bars together to make it easier to slide into the socket in the machine.



Crease/Perf Depth

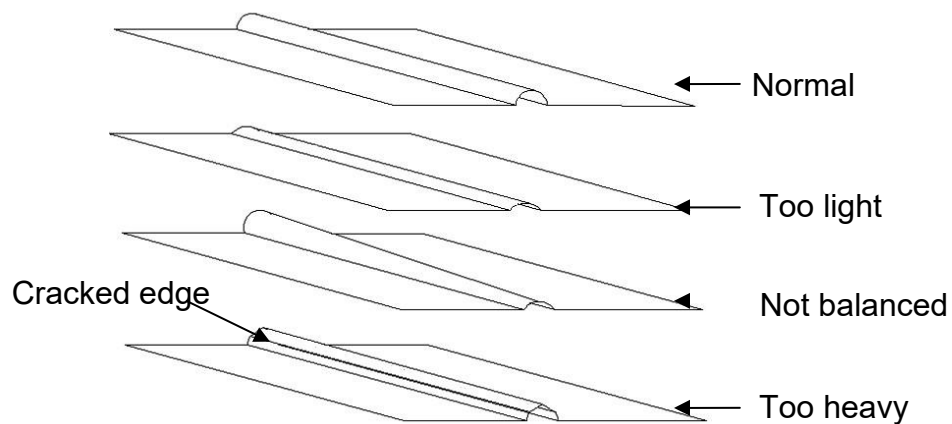
In some cases, the operator may choose to reduce or increase the pressure of the perf or crease depending on stocks. This adjustment will seldom have to be made unless the stock being run differs in paper weight between two extremes on a run to run basis. Generally, the factory setting will be sufficient for all jobs.

In the case of inconsistent perfining or creasing across the page:

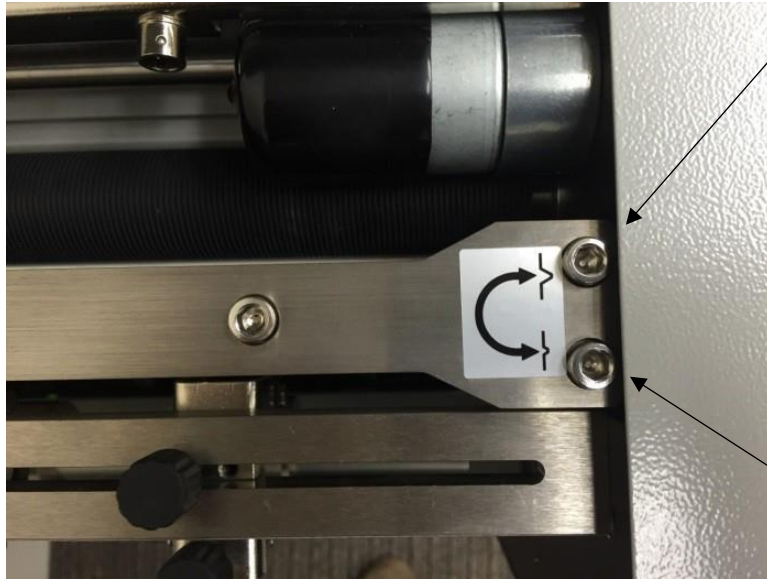
- Ensure first that there is nothing out of the ordinary with the rubber crash pad and crease/perf bar assembly. Adjustments should only be made as a last resort. If the crash pad is faulty please replace the pad before applying excessive pressure on the perfining unit.
- Do not make this adjustment if the pressure is even across the page.

- Crease depth is important to overall crease quality. Depth should be adjusted according to the paper thickness. Low depth will perform an unclear crease, but a high depth may cause cracking, or cause the system to take it as a double feed.

An adjustment is also needed if the crease depths are not balanced on both sides. To adjust the pressure of the crease/perf, please adjust the hex screws on either side of the bar evenly. By rotating either of the impression screws the operator can change the impact of the die accordingly.



Start by turning the screw $\frac{1}{8}^{\text{th}}$ or $\frac{1}{4}$ of a turn at a time until the impression is within the ideal zone.

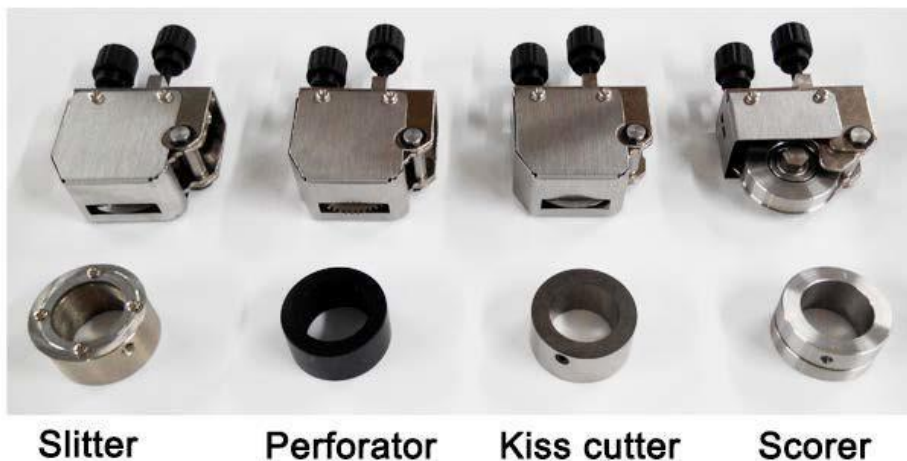


To increase pressure, both the top and the bottom screws need to be rotated clockwise.

To decrease pressure, both the top and the bottom screws need to be rotated counterclockwise.

Outfeed Accessories

The outfeed on the PT 335-A Plus Multi can handle various accessories including a strike perf feature. To set up the outfeed on the unit, use the thumbscrews and set screws to loosen or tighten the accessory holders and slide them along the bar. Ensure when setting up any sort of tool on the outfeed that there is a boss wheel underneath. These can be slid backwards, and forwards as needed by loosening/tightening the set screws.



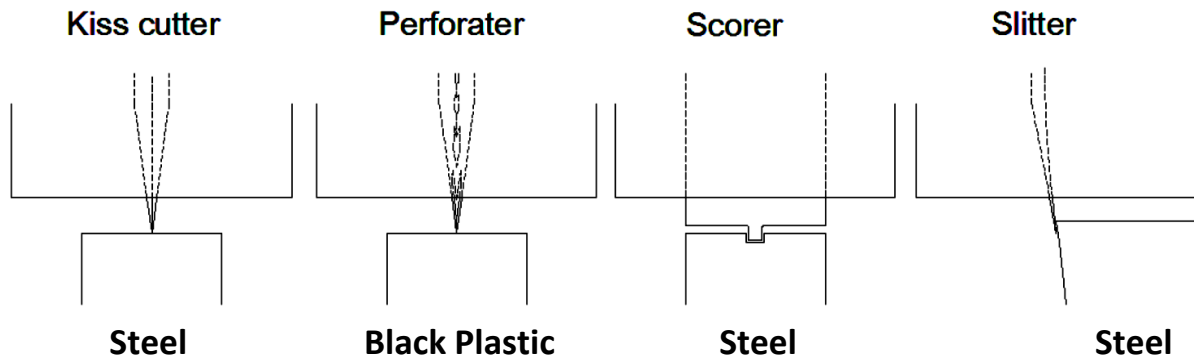
Slitter

Perforator

Kiss cutter

Scorer

There are four types of optional linear tools: **Scorer**, **Perforator**, **Kiss Cutter** and **Slitter**. Each tool is comprised of a blade and an anvil.



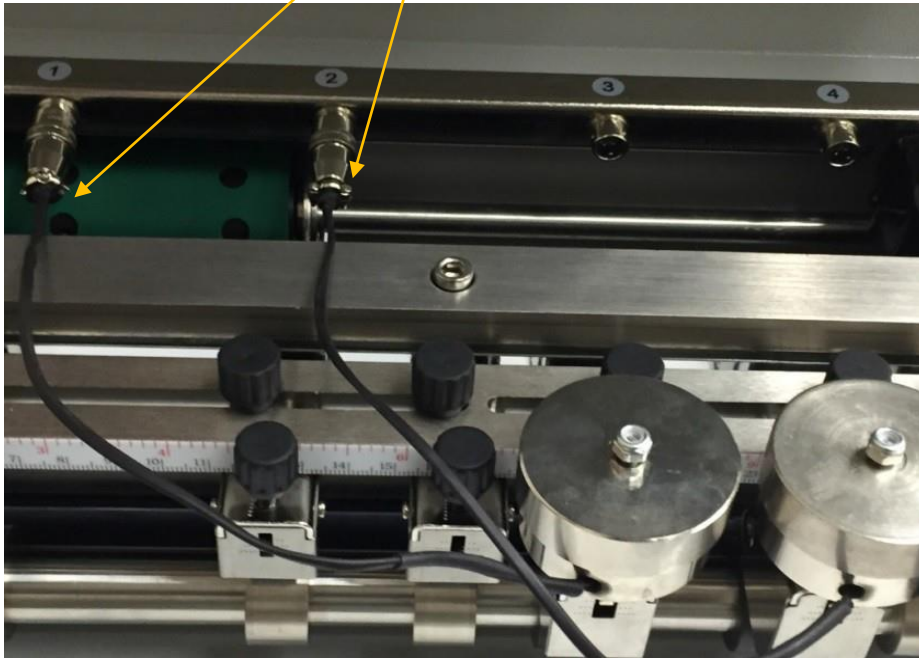
Installing tools on the tool mount shaft is as shown below:



Screw 1 fixes the tool onto the machine.

Screw 2 adjusts the depth of the tool, namely crease depth, perforate depth, kiss cutter depth, etc. This screw is particularly important to the kiss cutter adjustment.

To set up the strike perfs make certain that the perf unit is sitting on the outfeed bar and tightened in the position required. Connect the cable from the strike perf assembly into the corresponding slot that you require.



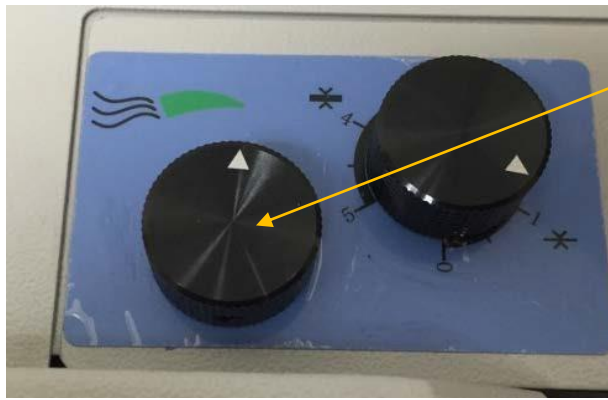
Please note that there are two pins in the connector piece for the strike perfs. Please connect gently by pressing the plug in straight. Forcing and/or twisting the plug into place may cause damage to the assembly.



The outfeed accessories will have to be manually aligned to perform a job. To aid in this process, the machine has a **“Check”** feature which will advance a piece of test paper. (This feature will be shown later in *Programming*).

Air Vent Control

By opening up the air vents at the front of the feeding unit more air will be dispatched to raise the stock.



The left dial on the control panel regulates the amount of air flow. By turning this dial, the vents on the feeding unit of the machine will open and close.

When running heavy stock, the air is advised to be open to raise the pile and ensure consistent feeding and sheet separation.

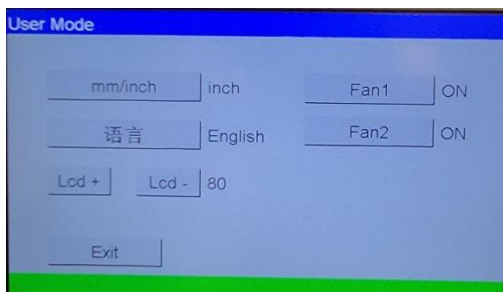




When running a light stock ensure the vents are more closed or the stock may blow around causing a miss feed or skewing.

Fan ON/OFF

To access fans in order to turn ON or OFF, go to the Home screen and select SET mode, then select USER mode.



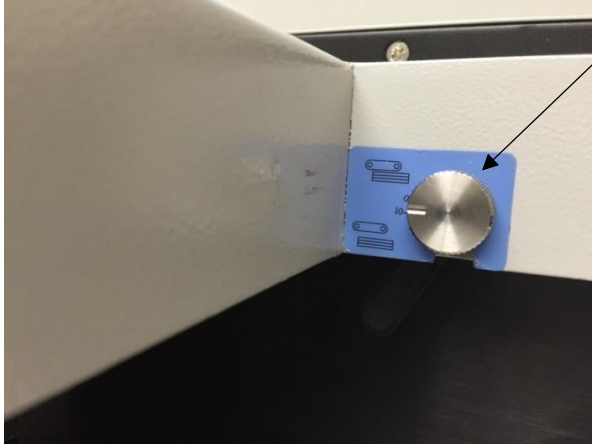
Select ON or OFF as required. When processing large or heavier sheets both fans should be set to ON. When jobs call for lighter stock, both fans will not be necessary.

Vacuum Control



Vacuum Control is located near the ON/OFF power switch. Turn dial CLOCKWISE to increase vacuum/suction to the feeder belt. This enables the Operator to feed heavier weight stock. Turning the dial COUNTER-CLOCKWISE will decrease vacuum/suction to the feeder belt, useful when processing lighter stock and will assist to reduce double feeding.

Pile Height Control



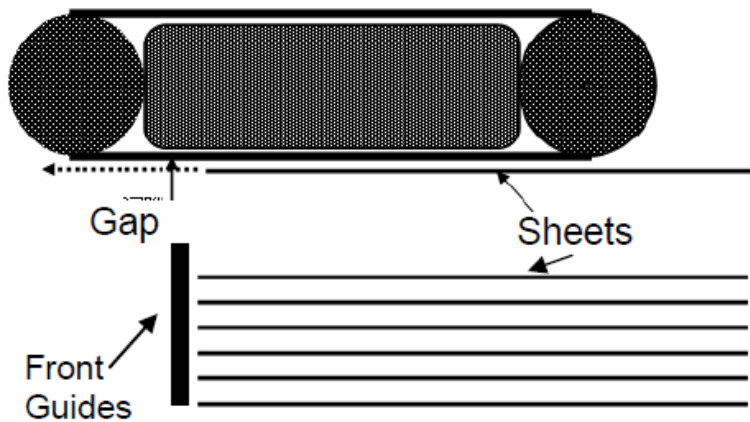
The press plate gap control knob located on the feeding units housing is designed to manipulate the size of the gap between the feeding belt and the stock. When using a lighter/thinner stock, the knob should be turned down to increase the gap between the feed belt and the sheets. If the stock is heavier, the pile height should be closer to the vacuum belt. Turn the knob up to 0.

Gap height adjustment will depend on the stock being used. If there seems to be a problem related to the feed belt not gripping the stock, then attempt to make this adjustment first.

Sheet Separator Adjustment

Turn the knob clockwise to reduce the gap if double feeding occurs. The front guides will move up to bridge the gap.

Turn the knob counterclockwise to increase the gap if paper is unable to go through. The front guides will move down to open the gap.



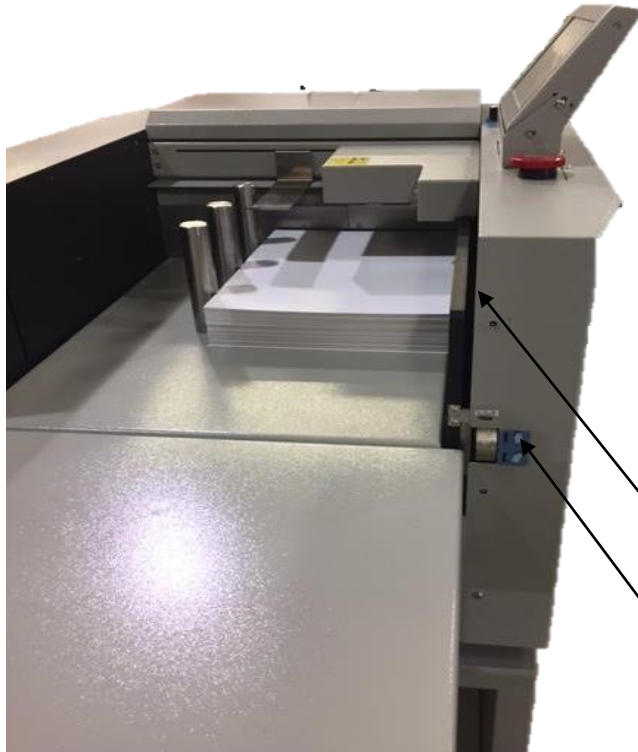
Skew Adjustment

If the crease/perforation is skewed, the angle of stationary side guide needs to be adjusted to change the crease angle.

Turning the skew adjustment wheel will skew your stock to account for any paper skew error.

As per the picture to the left, if the wheel is turned down, the stationary side plate will turn to the left, and vice versa. Normally, if the paper is perfectly square, no adjustment needs to be made.





IMPORTANT! When using skew adjustment, you must move paper back stop and magnetic side guides away from the edge of the paper stack. This will allow sheets to move as the skew wheel is turned. Return paper back stop and magnetic side guides to the edge of the paper stack. If skew adjustment is not set at the center, the lead edge of the paper pile should not be flush with the front edge of the feeder.

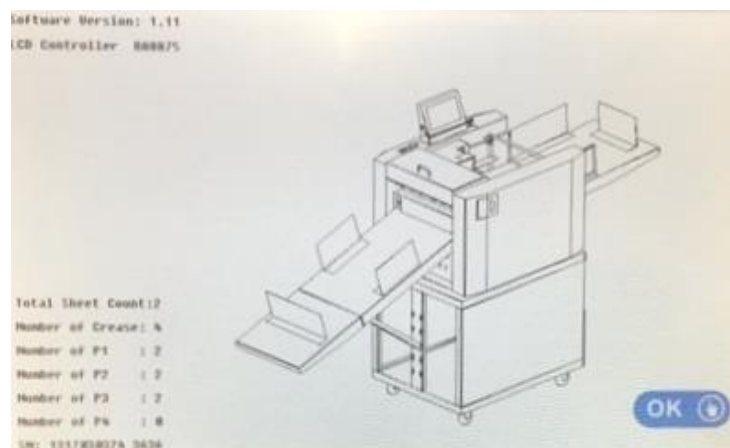
Fixed Side Plate

Skew Adjust Wheel

Programming

When the machine is switched on, the main screen will be displayed. This screen will give you some essential information about the machine and terms of its use. Press **OK** to continue to the home screen.

Note: After leaving the home screen, stock must be placed on the feed table. If there is no stock an error message will appear *"Please Set Paper"* before the machine can be used.



After pressing **OK** the next screen looks like the following. From this screen the operator can run a previously programmed job, run the last job, or program a new job.

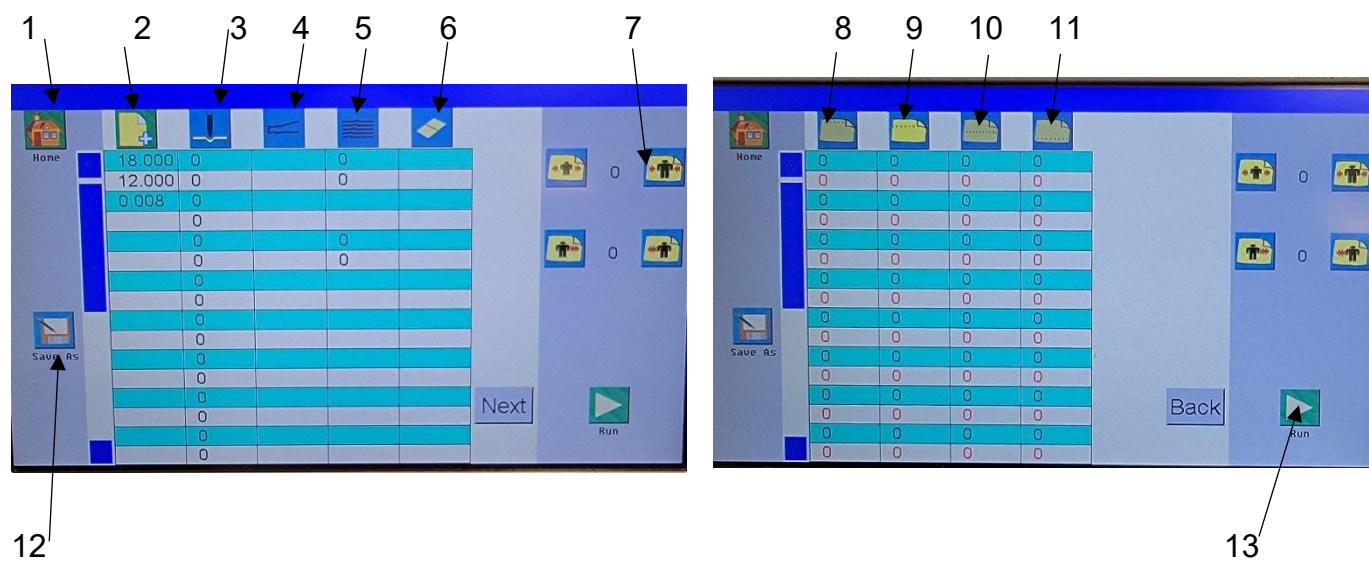


Edit	Job input/creator
Open	Opens saved jobs
Run	Runs last programmed job
Set	Access settings/service
Down	Lowens feed table to bottom position

By pressing **EDIT** you will be brought to the following screen (next page).

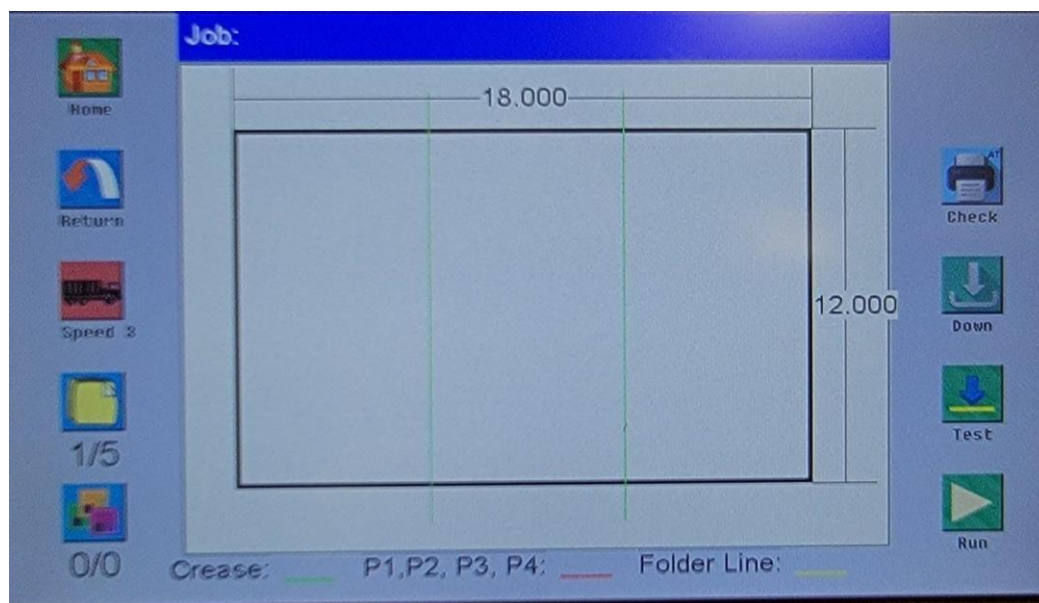
Notice that it has values already calibrated from the previous job. To run this job again just click on the **Run** button in the bottom right corner of the screen, or, modify by clicking on the icons along the top of the screen as shown below.

Use these icons to modify an existing job or remove them and program a new job manually following the steps provided below.



1	Home
2	Paper Size
3	Crease/Perf Position
4	Cover Hinge/Perfect Bind
5	Progressive Crease
6	Folder (Optional)
7	Shrink/Stretch adjustment
8	Strike Perf 1
9	Strike Perf 2
10	Strike Perf 3
11	Strike Perf 4
12	Save As
13	Run

Pressing **Run** will bring you to the main preview screen. This screen will be displayed when the machine is ready to run a programmed job.

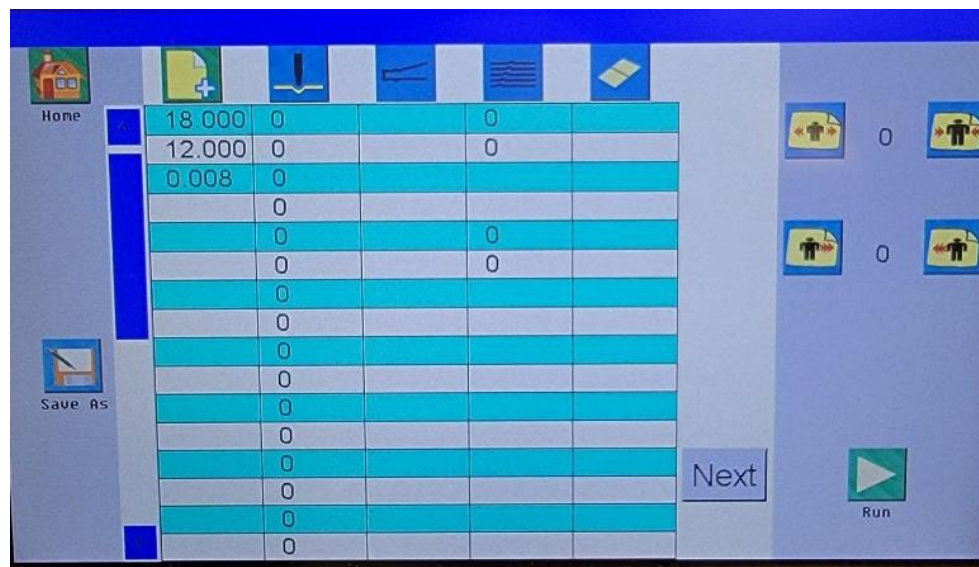


Home	Return to the home screen
Return	To data input screen
Speed	Speed
Count	Sheet Count, example shows #1 of 5 pages
Batch	Batch Count
Check	Feeds a sheet to align output
Down	Lowens the feed table
Test	Runs one sheet
Run	Begin running the job

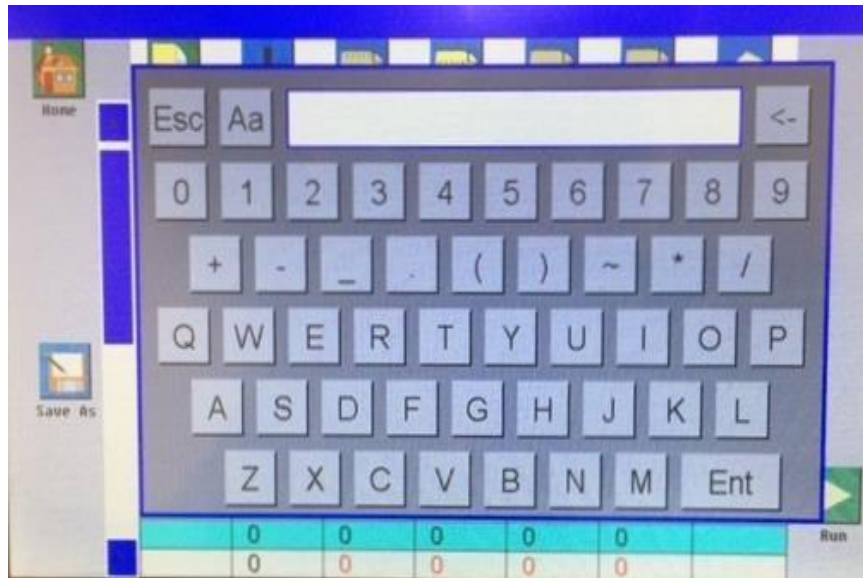
Job Set Up

If you want to edit the job, press the EDIT icon and follow the steps outlined below.

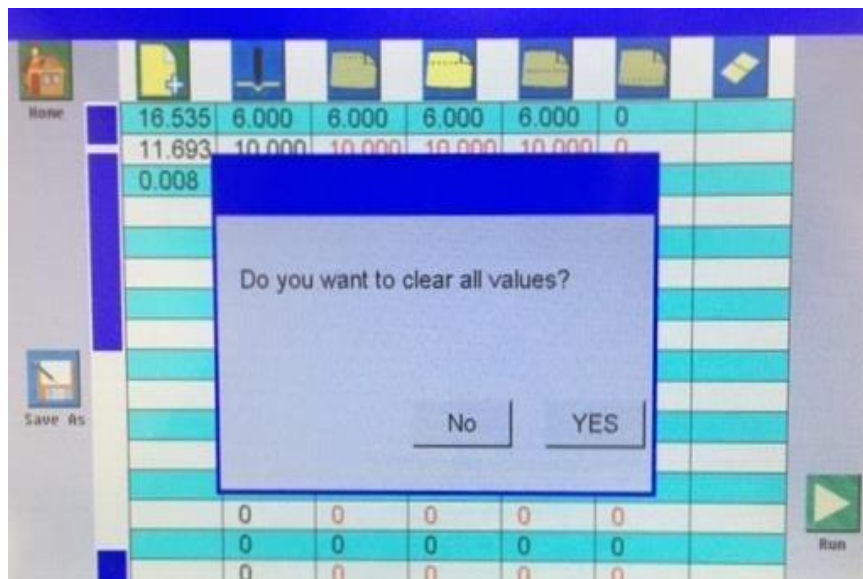
The following screens give the operator the ability to access the edit to screens for all the machine's functions. The operator may choose to edit values for paper size, crease/perf position, strike perfing, return home to open a saved job, save a job that has been programmed, or move forward to the run screen to perform a job once programming is finished.



To save a job, press the **Save As** icon on the left side of the screen. A touch keyboard will be displayed allowing the operator to name a job whatever they choose to recall it quickly for recurring runs.



When the operator chooses the **Paper Size** icon (first icon/first column of input) the machine will ask the user “Do you want to clear all values?” Choose **No** or **Yes**. This allows the operator to erase all the data and start programming a new job, or if they choose to retain the information already shown and make specific changes.



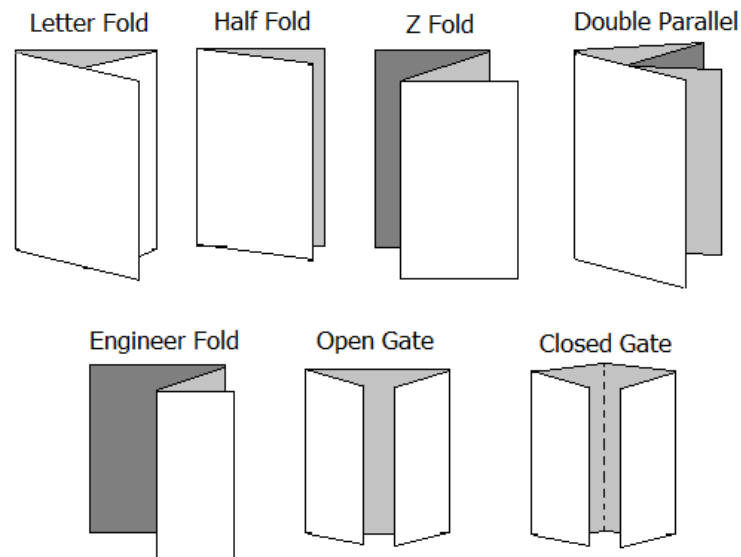
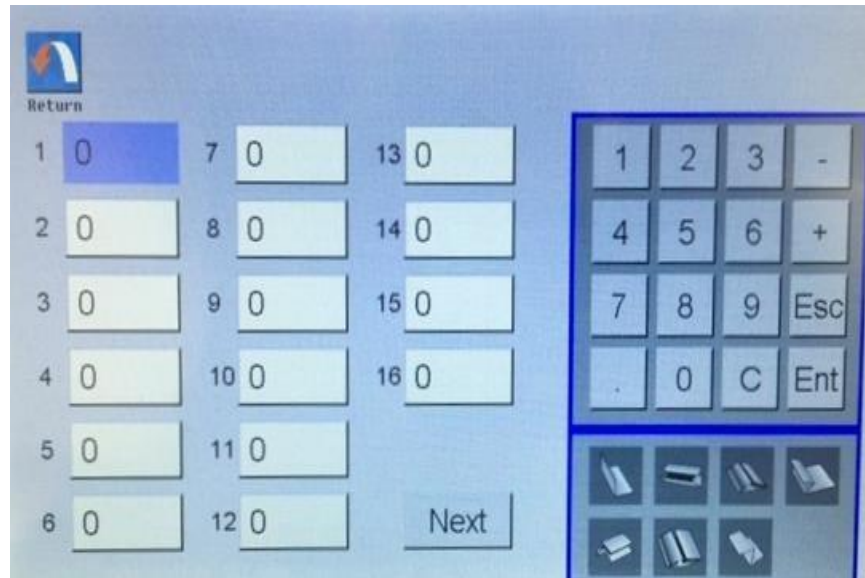
This screen (below) will appear next after clicking the **No** or **Yes** icon.

The screenshot shows a control interface with a 'Return' button in the top left. Below it are three input fields: 'Length: 11.000', 'Width: 8.500', and 'Thickness: 0.008'. To the right of the Length field is an 'Auto Measure' button. On the right side of the screen is a numeric keypad with buttons for digits 1-9, 0, a decimal point, and function keys like '-', '+', 'Esc', 'C', and 'Ent'. Below the numeric keypad are four buttons labeled 'Letter', 'Legal', 'Ledger', and 'Digital'.

The operator has the option to modify the paper size manually or use the auto measure function by pressing the **Auto Measure** button with a sheet loaded. This will send a sheet through the machine while it reads the lead and tail edge to fill in the length automatically for a precise measurement. Below the keypad is also a set of buttons with standard machine paper sizes for quick input.

Press **Return** to go back to the input screen and then choose the next icon over to change the crease/perf values.

Enter the cross perfing/creasing positions manually by entering the desired positions in inches or choose from a preset of typical jobs.



The auto job preset function will automatically fill in the crease/perf positions. Any modifications for folding can be made from the Run screen later or alter creasing positions if folding tightness is anticipated.

Next, enter the strike perf positions for each strike perf unit. The first line signifies where the perf will begin, the second when the perf will end.

To do a partial perf enter the starting and stopping point. If you want to make a full page enter the start and end of the page length, etc. Using the **=P1**, **=P2**, **=P3**, **=P4** buttons below the keypad the operator can copy the values which were inputted for one strike perf assembly and copy them to another strike perf. This will save time if multiple heads are performing the same perf job.

The image shows a control panel interface with a light blue background. At the top left is a logo with a stylized 'A' and the word 'Return' below it. The main area contains a grid of input fields for positions 1 through 8. Each position has two fields: a top field for the starting point and a bottom field for the ending point. All fields currently display '0'. To the right of this grid is a numeric keypad with buttons for digits 1-9, a decimal point, a zero, and function keys: '-', '+', 'Esc', and 'Ent'. Below the keypad are four buttons labeled '=P1', '=P2', '=P3', and '=P4'. At the bottom right of the grid is a 'Next' button.

Position	Start	End
1	0	0
2	0	0
3	0	0
4	0	0
5	0	0
6	0	0
7	0	0
8	0	0

Next

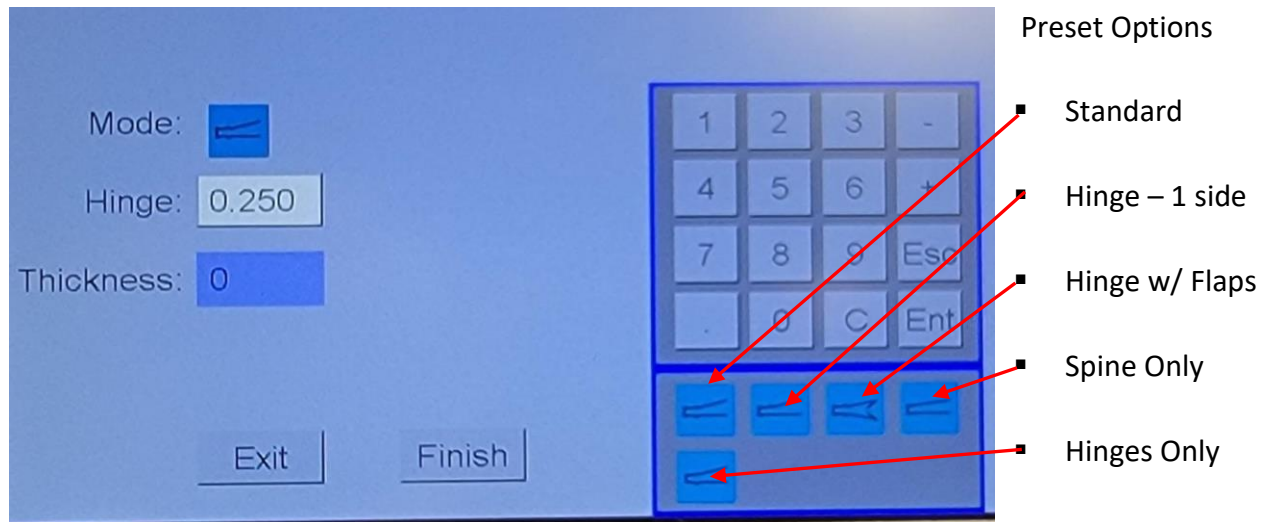
Keypad: 1, 2, 3, -, 4, 5, 6, +, 7, 8, 9, Esc, ., 0, C, Ent

Copy buttons: =P1, =P2, =P3, =P4

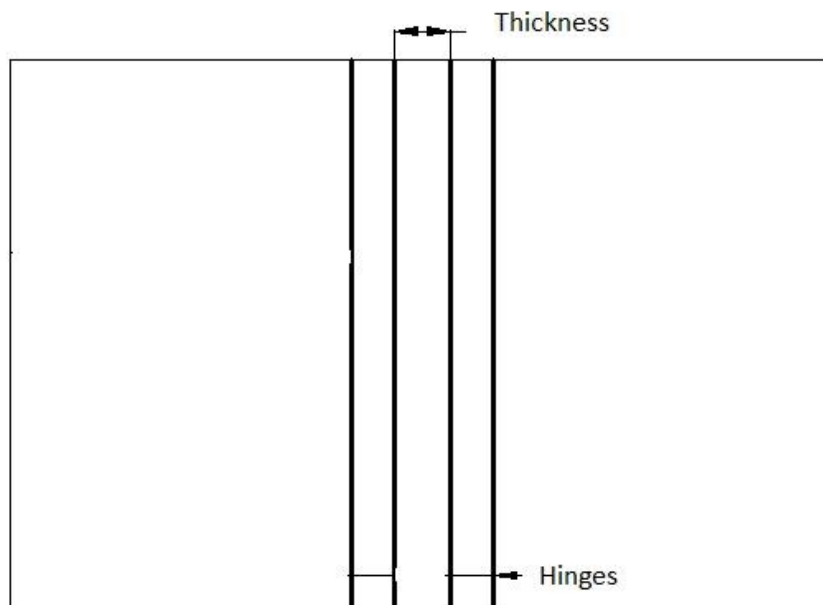
Once finished press **Return** and the job parameters will be shown for adjustment or review. If satisfied press **Run** to begin the job or run a test sheet to confirm.

Cover Hinge/Perfect Bind

This is a mode specifically for book cover making. The Operator inputs the hinge and thickness of the book and the machine will self-center the spine.

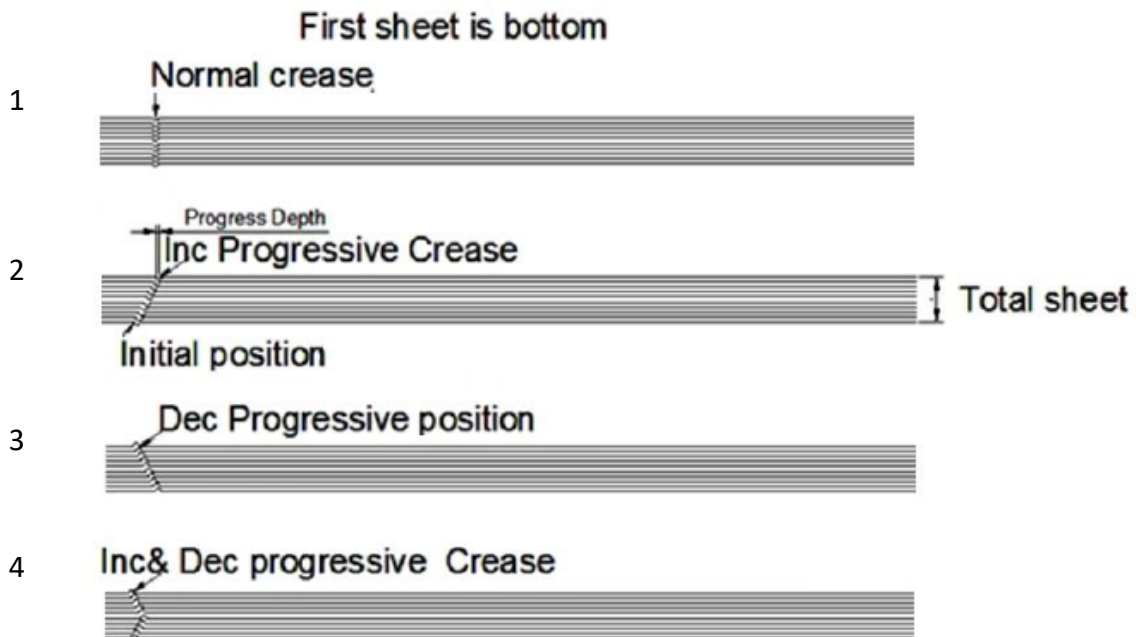
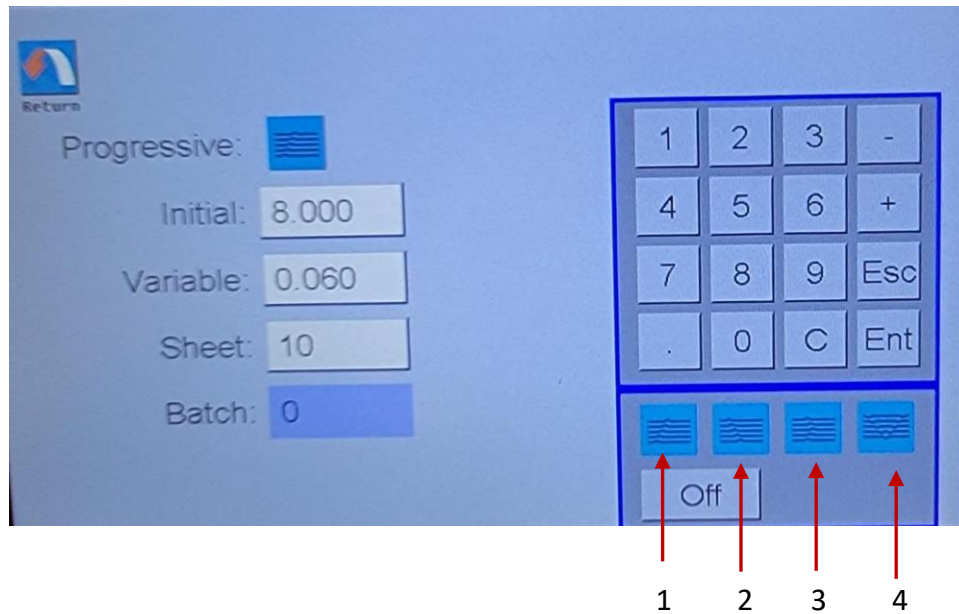


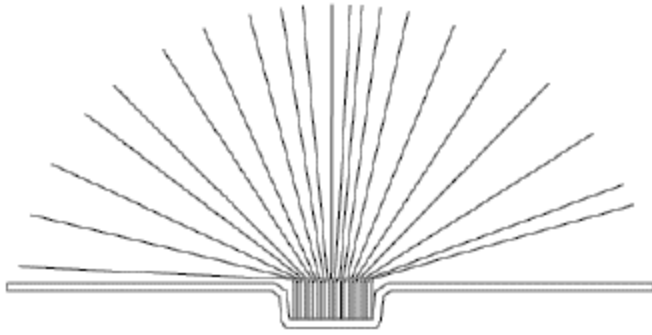
Example -



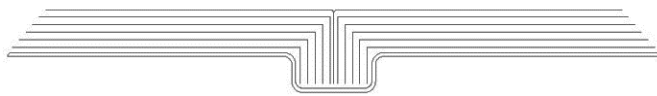
Progressive Crease

Select progressive creasing; 4 options as per below. The first initial crease is programmed, the number of sheets in the group and the variable (in the example below it is .06)



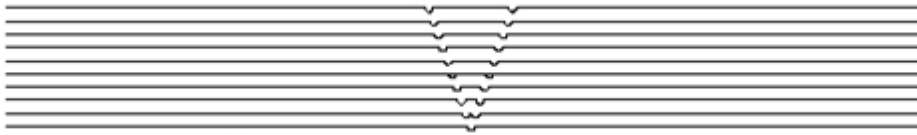


If all the sheets are creased on same position, the bounded book cannot be turned over to lay flat like this drawing.



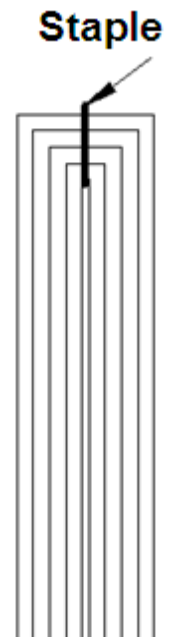
With 3 crease modes, even thick paper can be turned over very easily and lay flat after binding.

The illustration for V mode is as follows:



Define the first position and crease gap and the qty in one batch. The first page will be one line as reference and following sheets will be 2 creases that goes further successively to the reference line.

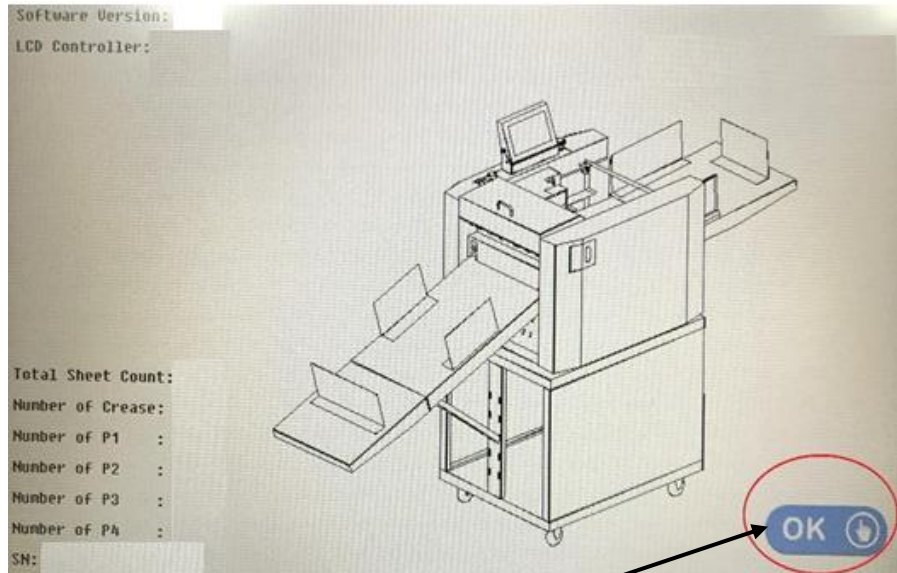
In this way the book can be folded with a square back after saddle stitched. The effect of the above 4 parameters is illustrated in the drawings above.



Troubleshooting

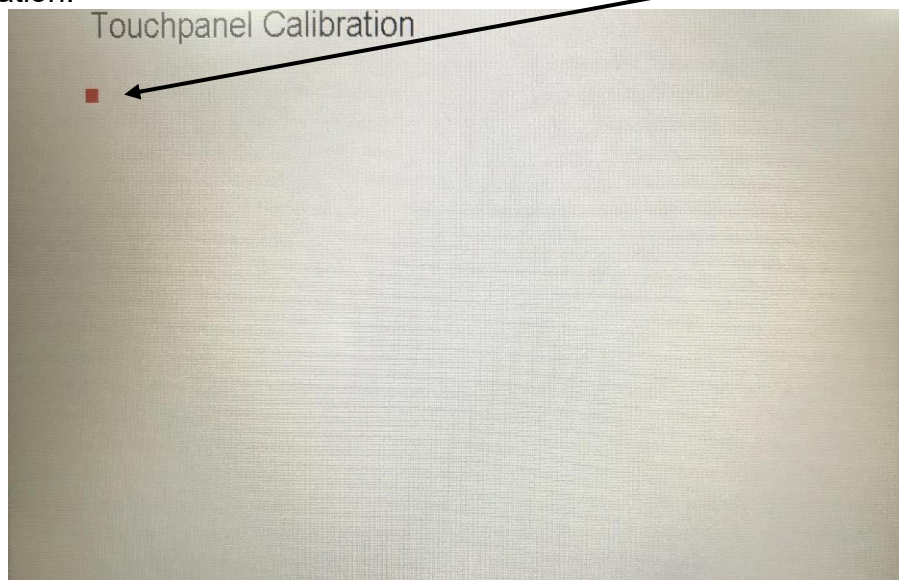
Touch Screen Calibration

Enter calibration mode by quickly touching the welcome screen 5 times.

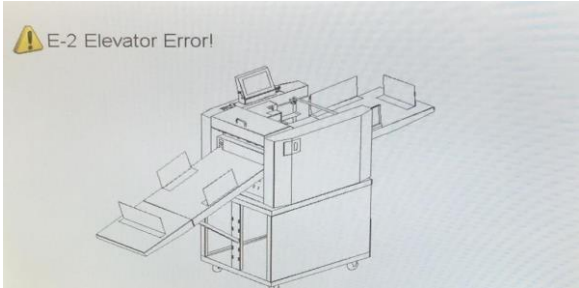
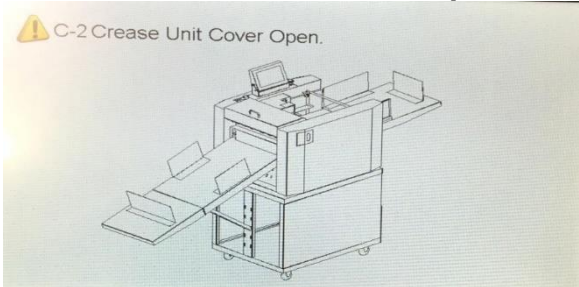


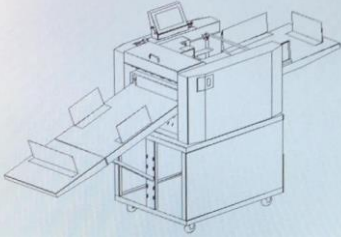
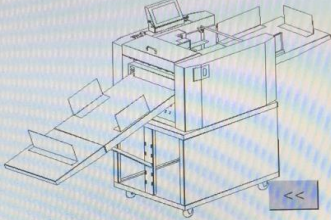
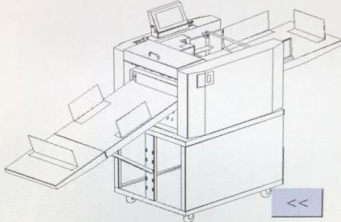
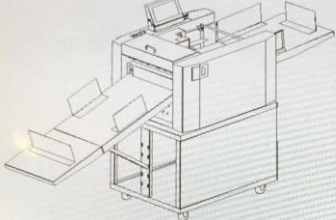
Do not touch the OK button.

The Touchpanel Calibration display will show up. Touch each red dot as they appear to finish calibration.



Error Codes

E-2 Elevator Error 	The PT 335A+ unit is equipped with a table height sensor as well as a lower safety switch. The elevator error will appear if one of these sensors is triggered. Power off the PT 335A+ unit then press and hold the green elevator button while powering the PT 335A+ back on. Continue to hold the green elevator button and press OK. While still holding the green elevator button, press the table down icon if the table is not at bottom position or press the Run icon (must have paper in feeder). If the table is up and the error occurs, this may indicate the paper stack is too high to feed. For correct feeding, the table needs room to rise. Power the unit on and off, remove some paper and try again. If the paper stack is not too large and the error keeps occurring, there may be an issue with the table height sensor. Contact your local GW dealer for technical assistance.
C-2 Crease Unit Cover Open 	The crease cover is open, or the limit switch is not activated when the cover is closed. Check for contact when closing the lid. If the lid is contacting the switch and the error still appears, the switch may require replacement. Contact

	your local GW dealer for technical assistance.
C-3 Crease Unit Side Cover Open  C-3 Crease Unit Side Cover Open. 	Ensure the side cover is fully closed. If the error still exists with the cover fully closed, there may be an issue with the switch. Contact your local GW dealer for technical assistance.
C-5 Paper Jam Near Infeed Roller  C-5 Paper Jam Near Infeed Roller. 	This problem occurs when the sensor near the infeed roller is covered. It may be covered by paper debris, check to see if this is what is causing the issue. If this is not the issue it may be an issue with the vacuum not pulling the paper enough.
C-6 Paper Jam Near Outfeed Roller  C-6 Paper Jam Near Outfeed Roller. 	Paper did not clear outfeed roller or outfeed sensor is blocked. Check for debris or paper jam. Clean sensor. This error can occur when the actual paper length does not match the length indicated in the programming. If the sensor is clear and there is no debris or jammed paper and the paper length is set correctly, contact your local GW dealer for technical assistance.
C-8 Please Set Paper  C-8 Please Set Paper. 	Feed table is empty. Placing paper on the feed table in front of the sensor and pressing return on the display will remove this message.

Maintenance

Proper maintenance will prolong the life and functionality of the PT 335A Plus Multi. These maintenance procedures are recommended with the assumption that the customer is completing daily and weekly cleaning. To ensure your machine is running at its best, please regularly perform the following:

- Using the GW Roller Cleaner, clean ink/toner dirt and debris from rollers. This should be done regularly to prevent build-up and protect the integrity of the rollers.
- Follow the rejuvenator cleaning with a wash of water and Simple Green: 1-part Simple Green to 30 parts water. Too much Simple Green will leave a soap residue.
- Lubricate the roller bushings with a 3 in 1 oil. This lubrication will ensure the paper is feeding properly and prevent the rollers from slowing or sticking.
- Clear paper dust from all sensors with compressed air.
- Clean all surfaces with a water-soluble cleaner.